Wind Turbine Power Output Prediction using Multiple Linear Regression.

Answer

The given dataset is,

Wind Speed (m/s)	Blade Angle (°)	Rotor Speed (RPM)	Power Output (kW)
5	11~~0~~	125	48
7	9	155	72
9	15	175	101
11	13	220	142
13	18	235	176

Here we can assign x and y to them. As we know y is dependent variable and x is independent variable.

The Power output is dependent over wind speed, Blade angle and Rotor Speed. Therefore we can rewrite the dataset as

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x_1	x_2	x_3	y
5	11	125	48
7	9	155	72
9	15	175	101
11	13	220	142
13	18	235	176

For manual Computation we use the formula

$$\theta = (X^T X)^{-1} (X^T y)$$

For that

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix}_{5\times4} \rightarrow \text{independent}$$

$$Y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}_{5\times1} \rightarrow \text{dependent}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix}_{4\times5}$$

$$(X^T X) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix}_{4\times5} \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix}_{5\times4}$$

$$= \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 920 & 12485 \\ 910 & 8760 & 12485 & 173900 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 76.901 & 18.214 & -3.004 & -1.104 \\ 18.214 & 4.586 & -0.724 & -0.274 \\ -3.004 & -0.724 & 0.160 & 0.040 \\ -1.1042 & -0.274 & 0.040 & 0.016 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

$$\theta = (X^T X)^{-1}(X^T Y) = \begin{bmatrix} -99.911 \\ 1.217 \\ 2.417 \\ 0.906 \end{bmatrix}$$

But we know $\theta = \begin{bmatrix} c \\ m_1 \\ m_2 \\ m_2 \end{bmatrix}$

Therefore the intercept $c = -99.911$

regression coefficients $m_1 = 1.217$

$m_2 = 2.417$

$m_3 = 0.906$

Now given in the question in order to find the Power output we use the model equation.

$$y = m_1x_1 + m_2x_2 + m_3x_3 + c$$

The given input to predict the answer of the power output,

x_1, wind speed = 9 m/s

x_2, Blade angle = 11°

x_3 = rotor speed = 100 RPM

$\therefore$ Power output, $Y = 1.217 \times 9 + 2.417 \times 11 + 0.906 \times 100 - 99.911$

$= \underline{28.208 \text{ kW}}$

Therefore 28.208 kW is the Power Output